TWISTING BY THE SIGN SHEAF PRESERVES QUASI-COHERENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The setup and notation of the twisting construction. Same chapter spaces-cohomology.tex, label lemma-twist (cited at statement line 1307). It fixes S, W, G, U and the character χ and defines the sheaf $\underline{\mathbf{Z}}(\chi)$ whose tensor product is being considered.
- Quasi-coherence is local for the etale topology, and $\underline{\mathbf{Z}}(\chi)$ is finite locally free. Same chapter spaces-cohomology.tex, the properties of $\underline{\mathbf{Z}}(\chi)$ established around the twisting construction (used at proof lines 1313-1317). The module structure is immediate, and the assertion may be checked etale locally, where tensoring with a finite locally free module preserves quasi-coherence.

Source reference. Original lemma 0727: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 1146–1146. Navigation pointer for \ref{lemma-twist} used inside the counted span (line 1307); the argument that consumes it is retained in the span.

Lemma 0.1. *With S, W, G, U, χ as in Lemma 0727. If $\mathcal{F}$ is a quasi-coherent $\mathcal{O}_U$-module, then so is $\mathcal{F} \otimes_{\mathbf{Z}} \underline{\mathbf{Z}}(\chi)$.*

Proof. The $\mathcal{O}_U$-module structure is clear. To check that $\mathcal{F} \otimes_{\mathbf{Z}} \underline{\mathbf{Z}}(\chi)$ is quasi-coherent it suffices to check étale locally. Hence the lemma follows as $\underline{\mathbf{Z}}(\chi)$ is finite locally free as a $\underline{\mathbf{Z}}$-module. □

Date: stacks-project:spaces-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.